Debian Zerbitzariaren Konfigurazio Gida: Peaje Sistema

1. Prestaketa eta SSH Sarbidea

Lehen urratsa zerbitzaria eguneratzea eta SSH zerbitzua instalatzea da. Honek Windows-eko CMD-tik komandoak kopiatu eta itsastea ahalbidetuko dizu. **Gogoratu bridge moduan jartzea VM-a.**

```
sudo apt update && sudo apt upgrade -y
sudo apt install openssh-server -y
sudo systemctl enable --now ssh
```

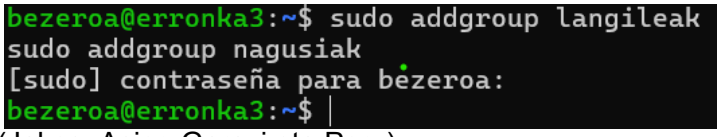
Zure IPa jakiteko Windows-etik konektatu ahal izateko:

```
sudo ip a
```

2. Erabiltzaileak eta Taldeak (Peajeko Langileak) Peajearen kudeaketarako bi

talde sortuko ditugu: **langileak** eta **nagusiak**.

```
sudo addgroup langileak
sudo addgroup nagusiak
```



Lau erabiltzaileak sortuko ditugu (Jakes, Asier, Garazi eta Peru):

```
sudo adduser jakes
sudo adduser asier
sudo adduser garazi
sudo adduser peru
```

```
bezeroa@erronka3:~$ sudo adduser jakes
sudo adduser asier
sudo adduser garazi
sudo adduser peru
Nueva contraseña:
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correctamente
Cambiando la información de usuario para jakes
Introduzca el nuevo valor, o pulse INTRO para usar el valor predeterminado
        Nombre completo []:
        Número de habitación []:
        Teléfono del trabajo []:
        Teléfono de casa []:
        Otro []:
Is the information correct? [Y/n] y
Nueva contraseña:
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correctamente
Cambiando la información de usuario para asier
Introduzca el nuevo valor, o pulse INTRO para usar el valor predeterminado
        Nombre completo []:
        Número de habitación []:
        Teléfono del trabajo []:
        Teléfono de casa []:
        Otro []:
Is the information correct? [Y/n] y
Nueva contraseña:
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correctamente
Cambiando la información de usuario para garazi
Introduzca el nuevo valor, o pulse INTRO para usar el valor predeterminado
        Nombre completo []:
        Número de habitación []:
        Teléfono del trabajo []:
        Teléfono de casa []:
        Otro []:
Is the information correct? [Y/n] y
Nueva contraseña:
Vuelva a escribir la nueva contraseña:
passwd: contraseña actualizada correctamente
Cambiando la información de usuario para peru
```

Erabiltzaile bakoitza dagokion taldean sartuko dugu:

sudo usermod -aG langileak jakes
sudo usermod -aG langileak asier
sudo usermod -aG nagusiak garazi
sudo usermod -aG nagusiak peru

```
bezeroa@erronka3:~$ sudo usermod -aG langileak jakes
sudo usermod -aG langileak asier
sudo usermod -aG nagusiak garazi
sudo usermod -aG nagusiak peru
bezeroa@erronka3:~$
```

3. Karpetak eta Baimenak (Peaje Dokumentazioa) Peajearen informazioa

gordetzeko direktorioak sortuko ditugu.

sudo mkdir -p /srv/peaje/diru_bilketa
sudo mkdir -p /srv/peaje/kamara_grabazioak

```
bezeroa@erronka3:~$ sudo mkdir -p /srv/peaje/diru_bilketa
sudo mkdir -p /srv/peaje/kamara_grabazioak
bezeroa@erronka3:~$
```

Baimenak esleitu: *diru_bilketa* nagusientzat bakarrik eta *kamara_grabazioak* taldearentzat.
sudo chown :nagusiak /srv/peaje/diru_bilketa
sudo chmod 770 /srv/peaje/diru_bilketa
sudo chown :langileak /srv/peaje/kamara_grabazioak
sudo chmod 775 /srv/peaje/kamara_grabazioak

```
bezeroa@erronka3:~$ sudo chown :nagusiak /srv/peaje/diru_bilketa
sudo chmod 770 /srv/peaje/diru_bilketa
sudo chown :langileak /srv/peaje/kamara_grabazioak
sudo chmod 775 /srv/peaje/kamara_grabazioak
bezeroa@erronka3:~$
```

4. Disko Kuotak

Erabiltzaileek diskoa bete ez dezaten (adibidez grabazioekin), mugak ezarriko ditugu.

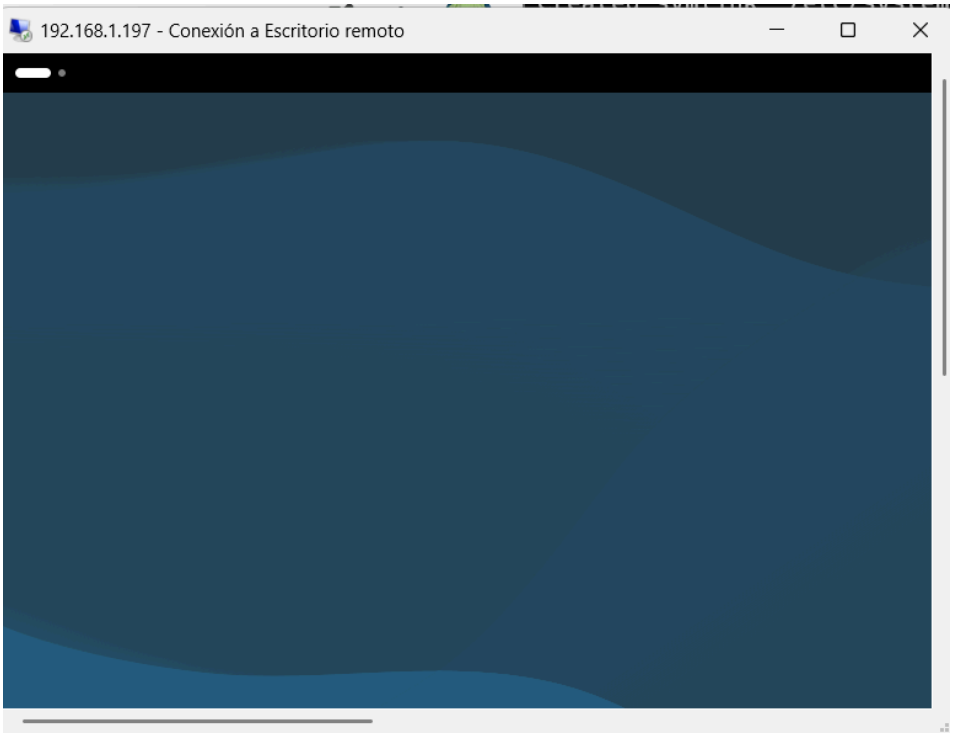
sudo apt install quota quotatool -y
sudo mount -o remount /
sudo quotacheck -cumg /
sudo quotaon -v / sudo edquota -u jakes

```
bezeroa@erronka3:~$ sudo apt install quota quotatool -y
sudo mount -o remount /
sudo quotacheck -cumg /
sudo quotaon -v /
sudo edquota -u jakes
Installing:
  quota  quotatool

Paquetes sugeridos:
  libnet-ldap-perl  rpcbind  default-mta  | mail-transport-agent

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 271 kB
  Space needed: 1.509 kB / 13,5 GB available

Des:1 http://deb.debian.org/debian trixie/main amd64 quota amd64 4.09-1+b1 [
251 kB]
Des:2 http://deb.debian.org/debian trixie/main amd64 quotatool amd64 1:1.6.5
-2 [20,4 kB]
Descargados 271 kB en 0s (1.456 kB/s)
Preconfigurando paquetes ...
Seleccionando el paquete quota previamente no seleccionado.
(Leyendo la base de datos ... 138972 ficheros o directorios instalados actua
lmente.)
Preparando para desempaquetar .../quota_4.09-1+b1_amd64.deb ...
Desempaquetando quota (4.09-1+b1) ...
Seleccionando el paquete quotatool previamente no seleccionado.
Preparando para desempaquetar .../quotatool_1%3a1.6.5-2_amd64.deb ...
Desempaquetando quotatool (1:1.6.5-2) ...
Configurando quotatool (1:1.6.5-2) ...
Configurando quota (4.09-1+b1) ...
Procesando disparadores para man-db (2.13.1-1) ...
quotacheck: Mountpoint (or device) / not found or has no quota enabled.
quotacheck: Cannot find filesystem to check or filesystem not mounted with q
uota option.
quotaon: Mountpoint (or device) / not found or has no quota enabled.
No filesystems with quota detected.
bezeroa@erronka3:~$
```

5. Ingurune Grafikoa eta Urruneko Mahaigaina (RDP) Zerbitzaria era grafikoan kontrolatzeko XFCE eta XRDP instalatuko dugu.

sudo apt install task-xfce-desktop -y
sudo apt install xrdp -y
sudo systemctl enable --now xrdp

```
bezeroa@erronka3:~$ sudo apt install task-xfce-desktop -y
sudo apt install xrdp -y
sudo systemctl enable --now xrdp
Installing:
  task-xfce-desktop

Installing dependencies:
  atril                          mate-polkit-common
  atril-common                   mousepad
  avahi-utils                    network-manager-applet
  exfalso                        parole
  exo-utils                      pavucontrol
  fonts-font-awesome             python3-feedparser
  fonts-lato                     python3-musicbrainzngs
  fonts-mathjax                  python3-mutagen
  gir1.2-gtksource-3.0           python3-pyasyncore
  gir1.2-handy-1                 python3-pyinotify
  gir1.2-keybinder-3.0           python3-sgmllib3k
  gir1.2-packagekitglib-1.0      quodlibet
  gnome-accessibility-themes     ristretto
  gnome-icon-theme               sphinx-rtd-theme-common
  gnome-themes-extra             synaptic
  gnome-themes-extra-data        system-config-printer
  gstreamer1.0-alsa              tango-icon-theme
  gtk2-engines-pixbuf            thunar
  libatrildocument3t64           thunar-archive-plugin
  libatrilview3t64               thunar-data
  libayatana-appindicator3-1     thunar-media-tags-plugin
  libayatana-ido3-0.4-0          thunar-volman
  libayatana-indicator3-7        tumbler
  libburn4t64                    tumbler-common
  libcaja-extension1             xfburn
  libdbus-glib-1-2               xfce4
  libexo-2-0                     xfce4-appfinder
  libexo-common                  xfce4-battery-plugin
  libgail-common                 xfce4-clipman
  libgail18t64                   xfce4-clipman-plugin
  libgarcon-1-0                  xfce4-cpufreq-plugin
```

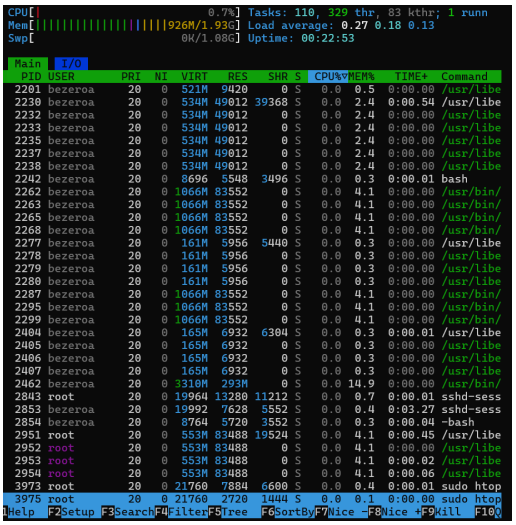


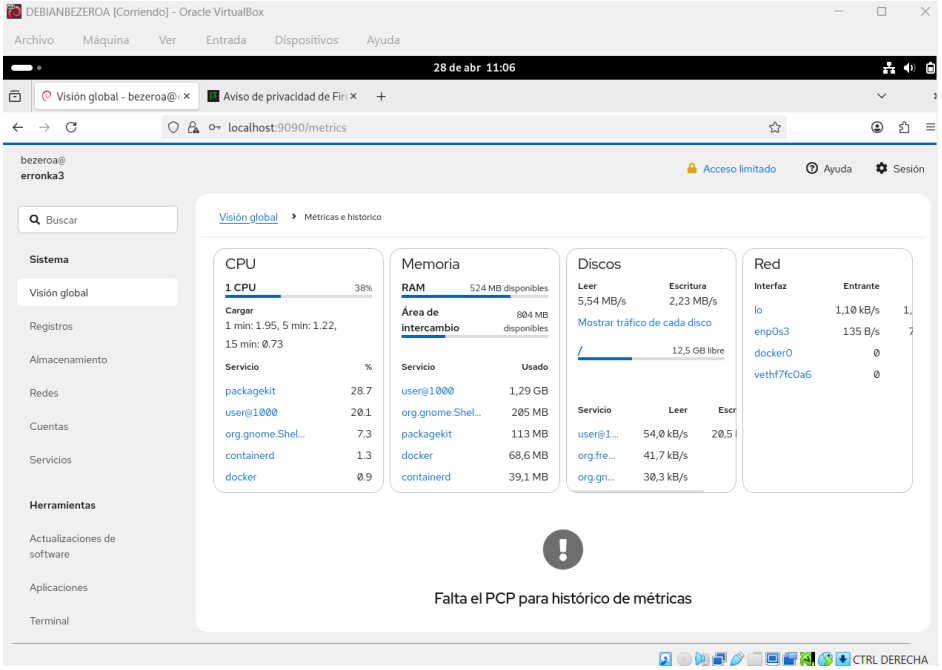
6. Cockpit eta Monitorizazioa

CPU, RAM eta sarea web bidez (9090 portuan) eta kontsolatik ikusteko.

sudo apt install cockpit -y
sudo free -m
sudo df -h
sudo apt install htop -y && sudo htop

```
bezeroa@erronka3:~$ sudo apt install cockpit -y
sudo free -m
sudo df -h
sudo apt install htop -y && sudo htop
Installing:
  cockpit

Installing dependencies:
  cockpit-bridge          cockpit-storaged  libpwquality-tools
  cockpit-networkmanager  cockpit-system
  cockpit-packagekit      cockpit-ws

Paquetes sugeridos:
  cockpit-doc         python3-pcp    udisks2-lvm2  lastlog2
  cockpit-sosreport   udisks2-btrfs  mdadm         sssd-dbus

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 0
  Download size: 7.328 kB
  Space needed: 9.201 kB / 13,5 GB available
```





7. Docker eta Rendimendua

Datuen analisia eta alertak kudeatzeko Netdata edukiontzi bat erabiliko dugu.

```
sudo apt install docker.io docker-compose -y
sudo docker run -d --name=peaje_monitor -p 19999:19999 netdata/netdata
```

```
bezeroa@erronka3:~$ sudo apt install docker.io docker-compose -y
sudo docker run -d --name=peaje_monitor -p 19999:19999 netdata/netdata
Installing:
  docker-compose  docker.io

Installing dependencies:
  containerd     liberror-perl              libsort-naturally-perl
  criu           libintl-perl               libterm-readkey-perl
  docker-buildx  libintl-xs-perl            needrestart
  docker-cli     libip4tc2                  patch
  git            libip6tc2                  python3-protobuf
  git-man        libmodule-find-perl        python3-pycriu
  iptables       libnet1                    runc
  libcompel1     libproc-processtable-perl  tini

Paquetes sugeridos:
  containernetworking-plugins  xfsprogs          git-cvs
  docker-doc                   zfs-fuse          git-mediawiki
  aufs-tools                   | zfsutils-linux  git-svn
  btrfs-progs                  git-doc           firewalld
  cgroupfs-mount               git-email         needrestart-session
  debootstrap                  git-gui           | libnotify-bin
  rinse                        gitk              ed
  rootlesskit                  gitweb            diffutils-doc

Summary:
  Upgrading: 0, Installing: 26, Removing: 0, Not Upgrading: 0
  Download size: 106 MB
  Space needed: 456 MB / 13,5 GB available
```

7.1. Docker Instalatu

Lehenik, edukiontziak kudeatzeko motorra instalatuko dugu peajeko zerbitzarian:

sudo apt update
sudo apt install docker.io docker-compose -y
sudo systemctl enable --now docker

```
bezeroa@erronka3:~$ sudo apt update
sudo apt install docker.io docker-compose -y
sudo systemctl enable --now docker
Obj:1 http://deb.debian.org/debian trixie InRelease
Des:2 http://deb.debian.org/debian trixie-updates InRelease [47,3 kB]
Des:3 http://security.debian.org/debian-security trixie-security InRel
43,4 kB]
Des:4 http://security.debian.org/debian-security trixie-security/main
s [132 kB]
Des:5 http://security.debian.org/debian-security trixie-security/main
Packages [127 kB]
Des:6 http://security.debian.org/debian-security trixie-security/main
ation-en [80,2 kB]
Descargados 430 kB en 0s (951 kB/s)
Leyendo lista de paquetes... 99%
```

7.2. Netdata Zabaldura (Monitorizazio Historikoa eta Web bidezkoa)

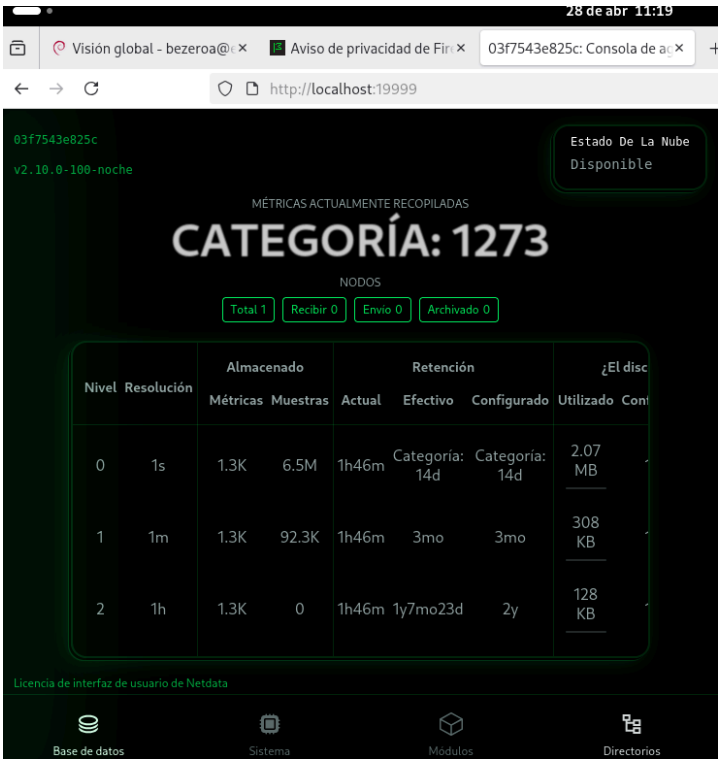
Netdata-k PUZ-aren, RAM-aren eta diskoaren erabilera segundoro aztertzen du. Datuak

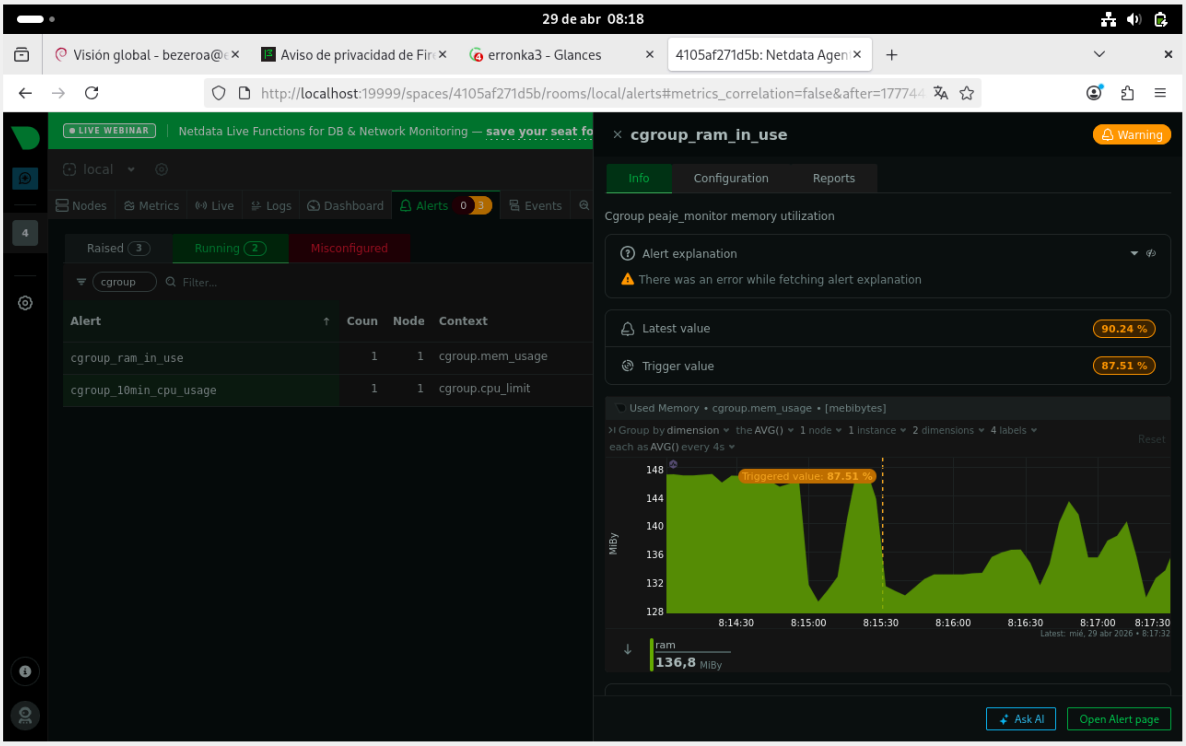
gordetzen ditu gero web bidez ikusteko eta alertak (adibidez, CPU %90etik gora badago) automatikoki konfiguratuta dakartza.

```
sudo docker run -d --name=peaje_monitor \
    -p 19999:19999 \
    -v netdataconfig:/etc/netdata \
    -v netdatalib:/var/lib/netdata \
    -v netdatacache:/var/cache/netdata \
    -v /var/run/docker.sock:/var/run/docker.sock:ro \
    -v /proc:/host/proc:ro \
    -v /sys:/host/sys:ro \
    -v /etc/os-release:/host/etc/os-release:ro \
    --restart unless-stopped \
    --cap-add SYS_PTRACE \
    --security-opt apparmor=unconfined \
    netdata/netdata
```

```
bezeroa@erronka3:~$ sudo docker run -d --name=peaje_monitor \
  -p 19999:19999 \
  --cpus=".5" \
  --memory="150m" \
  --memory-swap="150m" \
  -v netdataconfig:/etc/netdata \
  -v netdatalib:/var/lib/netdata \
  -v netdatacache:/var/cache/netdata \
  -v /var/run/docker.sock:/var/run/docker.sock:ro \
  -v /proc:/host/proc:ro \
  -v /sys:/host/sys:ro \
  -v /etc/os-release:/host/etc/os-release:ro \
  --restart unless-stopped \
  --cap-add SYS_PTRACE \
  --security-opt apparmor=unconfined \
  netdata/netdata
4105af271d5b92b45605e353c8e4981bad3c6c6eeab5742c399b4c21e00516fc
```

Oharra: *Behin hau eginda, joan zure nabigatzailera: http://zure_ip_debian:19999. Bertan peajearen errendimendu datu guztiak grafikoetan ikusiko dituzu.*





7.3. PUZ eta Memoria Mugak (Cgroups bidez)

Edukiontzi batek peajeko zerbitzariaren baliabide guztiak jan ez ditzan, mugak jar ditzakegu. Adibidez, prozesu bati PUZ-aren erdia eta 512MB RAM bakarrik erabiltzeko muga jartzeko:

Monitorizazio edukiontziari mugak ezartzea:

sudo docker update --cpus=".5" --memory="105m" --memory-swap="105m" peaje_monitor
sudo docker stats peaje_monitor --no-stream

```
bezeroa@erronka3:~$ sudo docker update --cpus=".5" --memory="105m" --memory-
swap="105m" peaje_monitor
peaje_monitor
bezeroa@erronka3:~$
```

```
bezeroa@erronka3:~$ sudo docker stats peaje_monitor --no-stream
CONTAINER ID   NAME            CPU %     MEM USAGE / LIMIT   MEM %     NET I
/O            BLOCK I/O        PIDS
03f7543e825c   peaje_monitor   5.35%     90.45MiB / 105MiB   86.14%    22.7k
B / 39.6kB   111MB / 49.5MB   99
bezeroa@erronka3:~$
```

7.4. Disko Gogorraren Mugak (LVM edo Quota bidez)

Peajeko grabazioek diskoa bete ez dezaten, aurreko atalean (4. atala) konfiguratu ditugun ***disko kuotak*** *dira mugarik eraginkorrenak. Erabiltzaile bakoitzak ezingo du ezarritako muga baino gehiago idatzi.*

Egiaztatu kuoten egoera peajean:

sudo repquota -s /

```
bezeroa@erronka3:~$ sudo repquota -s /
*** Report for user quotas on device /dev/sda1
Block grace time: 7days; Inode grace time: 7days
                        Space limits                File limits
User            used    soft    hard  grace    used  soft  hard  grace
----------------------------------------------------------------------
root      --   6863M      0K      0K           206k     0     0
man       --   1856K      0K      0K            183     0     0
lp        --      0K      0K      0K              1     0     0
_apt      --     20K      0K      0K              5     0     0
systemd-network --     20K      0K      0K              5     0     0
dhcpcd    --      8K      0K      0K              2     0     0
tss       --      4K      0K      0K              1     0     0
systemd-timesync --      4K      0K      0K              2     0     0

dnsmasq   --     16K      0K      0K              4     0     0
speech-dispatcher --      4K      0K      0K              1     0     0

fwupd-refresh --      4K      0K      0K              1     0     0
geoclue   --      4K      0K      0K              1     0     0
gnome-remote-desktop --      8K      0K      0K              2     0     0

saned     --      4K      0K      0K              1     0     0
colord    --     64K      0K      0K              7     0     0
Debian-gdm --   3668K      0K      0K            188     0     0
bezeroa   --    163M      0K      0K            794     0     0
jakes     --     24K      0K      0K              6     0     0
asier     --     24K      0K      0K              6     0     0
garazi    --     24K      0K      0K              6     0     0
peru      --     24K      0K      0K              6     0     0
lightdm   --      4K      0K      0K              1     0     0
xrdp      --      8K      0K      0K              2     0     0
#201      --  17236K      0K      0K            743     0     0


bezeroa@erronka3:~$
```

7.5. Alertak Konfiguratzea

Netdata-k mezuak bidal ditzake (Telegram, Email edo Discord bidez) PUZ-a edo RAM-a mugetara iristen denean. Konfigurazioa fitxategi honetan aldatzen da:

Alerta fitxategia editatzeko (edukiontzi barruan):
sudo docker exec -it peaje_monitor edit-config health.d/cpu.conf

```
        to: silent

## Windows

   template: 10min_cpu_usage
         on: system.cpu
      class: Utilization
       type: System
  component: CPU
host labels: _os=windows
     lookup: average -10m unaligned of irq,user,system,dpc
      units: %
      every: 1m
       warn: $this > (($status >= $WARNING)  ? (75) : (85))
       crit: $this > (($status == $CRITICAL) ? (85) : (95))
      delay: up 1m down 15m multiplier 1.5 max 1h
    summary: System CPU utilization
       info: Average CPU utilization over the last 10 minutes
        to: sysadmin

## FreeBSD
   template: 10min_cpu_usage
         on: system.cpu
      class: Utilization
       type: System
  component: CPU
host labels: _os=freebsd
     lookup: average -10m unaligned of user,system,interrupt
      units: %
      every: 1m
       warn: $this > (($status >= $WARNING)  ? (75) : (85))
       crit: $this > (($status == $CRITICAL) ? (85) : (95))
      delay: down 15m multiplier 1.5 max 1h
    summary: System CPU utilization
       info: Average CPU utilization over the last 10 minutes (excluding nic
e)
        to: sysadmin
                                                                  83,21         Bot
```

Web bidezko bistaratzea:

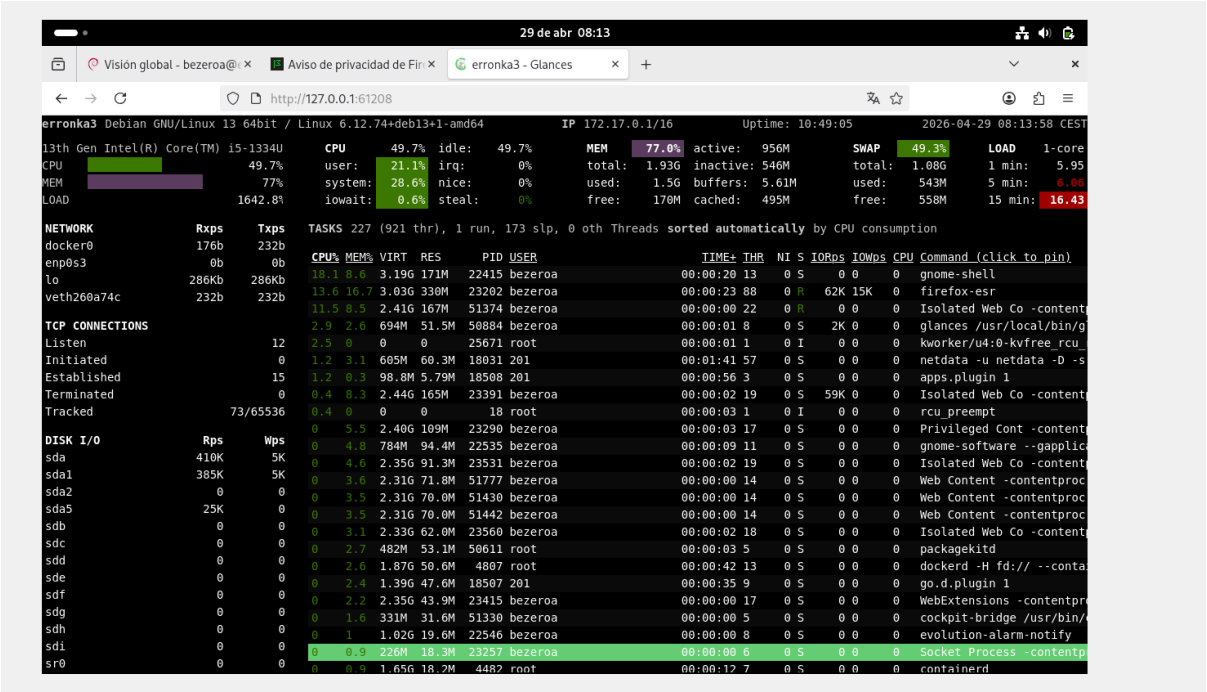
sudo apt install python3-pip -y

```
bezeroa@erronka3:~$ sudo apt install python3-pip -y
Installing:
  python3-pip

Installing dependencies:
  binutils                  g++-x86-64-linux-gnu       libbinutils    libfakeroot             libquadmath0      python3-packaging
  binutils-common           gcc                        libc-dev-bin   libfile-fcntllock-perl  libsframe1        python3-wheel
  binutils-x86-64-linux-gnu gcc-14                     libc6-dev      libgcc-14-dev           libstdc++-14-dev  python3.13-dev
  build-essential           gcc-14-x86-64-linux-gnu    libcc1-0       libgprofng0             libtsan2          rpcsvc-proto
  dpkg-dev                  gcc-x86-64-linux-gnu       libcrypt-dev   libhwasan0              libubsan1         zlib1g-dev
  fakeroot                  libalgorithm-diff-perl     libctf-nobfd0  libitm1                 linux-libc-dev
  g++                       libalgorithm-diff-xs-perl  libctf0        liblsan0                make
  g++-14                    libalgorithm-merge-perl    libdpkg-perl   libpython3-dev          manpages-dev
  g++-14-x86-64-linux-gnu   libasan8                   libexpat1-dev  libpython3.13-dev       python3-dev

Paquetes sugeridos:
  binutils-doc   debian-keyring            g++-14-multilib  autoconf  flex   gcc-doc          gdb-x86-64-linux-gnu  bzr              python3-setuptools
  gprofng-gui    debian-tag2upload-keyring gcc-14-doc       automake  bison  gcc-14-multilib  libc-devtools         libstdc++-14-doc
  binutils-gold  g++-multilib              gcc-multilib     libtool   gdb    gcc-14-locales   glibc-doc             make-doc

Summary:
  Upgrading: 0, Installing: 51, Removing: 0, Not Upgrading: 6
  Download size: 72,0 MB
  Space needed: 289 MB / 11,3 GB available
```

sudo pip3 install "glances[web]" --break-system-packages

```
bezeroa@erronka3:~$ sudo pip3 install "glances[web]" --break-system-packages
Collecting glances[web]
  Downloading glances-4.5.4-py3-none-any.whl.metadata (16 kB)
Collecting defusedxml (from glances[web])
  Downloading defusedxml-0.7.1-py2.py3-none-any.whl.metadata (32 kB)
Requirement already satisfied: packaging in /usr/lib/python3/dist-packages (from glances[web]) (25.0)
Requirement already satisfied: psutil>=5.6.7 in /usr/lib/python3/dist-packages (from glances[web]) (7.0.0)
Collecting shtab (from glances[web])
  Downloading shtab-1.8.0-py3-none-any.whl.metadata (7.3 kB)
Collecting jinja2 (from glances[web])
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting pyinstrument>=5.1.2 (from glances[web])
  Downloading pyinstrument-5.1.2-cp313-cp313-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (26 kB)
Collecting fastapi>=0.82.0 (from glances[web])
```

glances -w &



8. Sarean zeharreko Inplementazioa (PXE) Peajeko beste bezeroak saretik instalatzeko sistema.

sudo apt install dnsmasq -y

```
bezeroa@erronka3:~$ sudo apt install dnsmasq -y
Installing:
  dnsmasq

Paquetes sugeridos:
  resolvconf

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 69,5 kB
  Space needed: 159 kB / 11,9 GB available

Des:1 http://deb.debian.org/debian trixie/main amd64 dnsmasq all 2.91-1 [69,
5 kB]
Descargados 69,5 kB en 0s (528 kB/s)
Seleccionando el paquete dnsmasq previamente no seleccionado.
(Leyendo la base de datos ... 142240 ficheros o directorios instalados actua
lmente.)
Preparando para desempaquetar .../dnsmasq_2.91-1_all.deb ...
Desempaquetando dnsmasq (2.91-1) ...
Configurando dnsmasq (2.91-1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/dnsmasq.service
' → '/usr/lib/systemd/system/dnsmasq.service'.
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
bezeroa@erronka3:~$
```